

# Pulmonary gas exchange is reduced by the cardiovascular diving response in resting humans

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## Abstract

The diving response reduces the pulmonary O<sub>2</sub> uptake in exercising humans, but it has been debated whether this effect is present at rest. Therefore, respiratory and cardiovascular responses were recorded in 16 resting subjects, performing apnea in air and apnea with face immersion in cold water (10 °C). Duration of apneas were predetermined to be identical in both conditions (average: 145 s) and based on individual maximal capacity (average: 184 s). Compared to apnea in air, an augmented diving response was elicited by apnea with face immersion. The O<sub>2</sub> uptake from the lungs was reduced compared to the resting eupneic control (4.6 ml min<sup>-1</sup> kg<sup>-1</sup>), during apnea in air (3.6 ml min<sup>-1</sup> kg<sup>-1</sup>) and even more so during apnea with face immersion (3.4 ml min<sup>-1</sup> kg<sup>-1</sup>). We conclude that the cardiovascular adjustments of the diving response reduces pulmonary gas exchange in resting humans, allowing longer apneas by preserving the lungs' O<sub>2</sub> store for use by vital organs.  
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## 1. Introduction

The human cardiovascular diving response is characterized by bradycardia, reduced cardiac output, peripheral vasoconstriction and systemic hypertension (Gooden, 1994; Foster and Sheel, 2005). The blood flow is redistributed towards the vital organs during apnea, a notion that is supported by the observations of reduced peripheral blood flow together with increased blood flow in the carotid artery (Heistad et al., 1968; Sterba and Lundgren, 1988; Jiang et al., 1994; Pan et al., 1997). Both the cardiac and vascular responses are initiated by apnea and augmented by face immersion in cold water (Gooden, 1994; Marsh et al., 1995). In addition, splenic contraction is initiated during apnea, elevating circulating hematocrit (Schagatay et al., 2001; Bakovic et al., 2003). All of these mechanisms may serve to promote apneic duration.

In natural divers such as Weddell seals, the cardiovascular adjustments of the diving response is known to contribute to prolonged breath-hold diving (Butler and Jones, 1997). Of importance is a temporary shift towards increased anaerobiosis,

brought about by the reduced peripheral O<sub>2</sub> delivery due to the circulatory changes. While the diving response of some diving mammals thus has been shown to be O<sub>2</sub> conserving, the effect of the human diving response has been questioned. However, Ferretti et al. (1991) found that three elite breath-hold divers from the same family, but not untrained control subjects, reduced their alveolar gas exchange during resting apnea compared to the eupneic control level. In a larger group of subjects performing apneas in air and apneas with face immersion of equal duration during rest, Andersson and Schagatay (1998) found that the more pronounced diving response during apneas with face immersion was associated with an attenuation of the arterial desaturation. Since then, more studies have shown attenuating effects of the diving response on the arterial desaturation and alveolar gas exchange, at least in exercising subjects (Lindholm et al., 1999; Andersson et al., 2002, 2004; Lindholm and Linnarsson, 2002). In addition, during apneas combined with low-intensity exercise, a reduced pulmonary O<sub>2</sub> uptake was associated with blood lactic acid accumulation (Ferretti et al., 1991; Andersson et al., 2004). Reduced peripheral O<sub>2</sub> delivery, due to the diving response, probably causes a shift towards more anaerobiosis during apnea in exercising subjects. Thus, while the human diving response's oxygen-conserving effect has been shown in exercising subjects, the effect has not been conclusively shown and even doubted to

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be demonstrable in resting subjects (Hong et al., 1971; Lin, 1988; Lindholm et al., 1999). However, some studies have presented data indicative of an oxygen-conserving effect of the diving response in resting humans (Ferretti et al., 1991; Andersson and Schagatay, 1998; Schagatay and Andersson, 1998).

The pulmonary gas exchange in resting humans during apneas of the same duration either with the face in air or immersed in cold water have not been studied previously. Nevertheless, many apneic events in humans are associated with rest or very low levels of physical activity, e.g. sleep apnea and competitive “static apnea”. Therefore, we wanted to study pulmonary gas exchange, plasma lactate concentration, and cardiovascular responses in resting subjects during apneas without and with face immersion in cold water to test the hypotheses that the pulmonary  $O_2$  uptake would be temporarily reduced and plasma lactate concentration increased during apnea compared to eupneic control. We also hypothesized that a more pronounced diving response during apneas with face immersion would be associated with a lower pulmonary  $O_2$  uptake compared to during apneas with the face in air.

## 2. Methods

The study was conducted in accordance with the Declaration of Helsinki and was approved by the local research ethics committee. After receiving a description of the procedures and an explanation of potential risks involved, all subjects gave their written, informed consent.

### 2.1. Subjects

A group of 16 healthy, male subjects volunteered for the study. Their mean age was 24 year (range: 18–33 year), height 185 cm (175–196 cm), weight 81 kg (72–98 kg), vital capacity in the prone position 5.9 l (4.9–6.6 l), and residual volume 1.8 l (1.3–2.8 l). They were active breath-hold divers training breath-hold diving at least 2 h/week to daily ( $n = 9$ ), or recruited among subjects having performed relatively long apneas (>2.5 min) in earlier experiments at our laboratory. In addition to their breath-hold diving training, their physical training averaged 4 h/week (0–15 h/week). All subjects reported at the laboratory after at least 2 h without any heavy meal or caffeine-containing beverages.

### 2.2. Experimental protocol

During the entire experiment, the subject was lying as relaxed and comfortable as possible in the prone position on a mattress. The subject's head rested on a removable pillow on top of a container used for face immersions. The arms rested on both sides of the container at the level of the heart. During the experiments, water temperature was maintained at 10–11 °C and ambient air temperature at 23–25 °C.

The subject's vital capacity was measured. After at least 10 min of horizontal rest, when stable cardiovascular data were observed, recordings began. After 5 min of control recording, the subject performed a total of six apneas, alternating between

apnea with the face in air and apnea with face immersion at a weighted starting order. These conditions will hereafter be referred to as “apneas in air” and “apneas with face immersion”. To be able to determine each subject's maximal breath-holding time, the two first apneas were performed to the conventional breaking point (Lin et al., 1974) and spaced by 5 min of rest. Ten minutes after the second maximal duration apnea, the subject performed a series of four apneas with predetermined, sub-maximal breath-holding times, all spaced by 5 min of rest, i.e., the last four apneas were of identical duration, based on individual maximal capacity. Typically, after consulting the subject, the sub-maximal breath-holding time was set to be approximately 20 s shorter than the shorter of the two maximal attempts.

Before each apnea, without prior voluntary hyperventilation the subject exhaled to his residual volume through the mouthpiece of a spirometry system and inhaled, from a rubber bag, a volume of air corresponding to 90% of the individual prone vital capacity. During apneas in air, the subject held the face directly above the water surface, and during apneas with face immersion, the entire face, including the chin and forehead, was immersed. This assured the same body movements and position during all apneas. After countdown by the experimenter, the apneas of sub-maximal duration were terminated with a maximal exhalation through the mouthpiece. Five minutes after the last apnea, the residual volume was measured with a nitrogen-dilution technique (Rahn et al., 1949).

### 2.3. Measurements

Before the test, an electrocardiogram was recorded and checked for anomalies (Cardisun 501, Fukuda ME Kogoyo Co., Tokyo, Japan) and the resting arterial blood pressure was measured (average: 133/74 mmHg). The vital capacity was measured with a spirometer (Micro Plus, Micro Medical Ltd., Rochester, England). During the experiment, respiratory flow as well as expiratory  $O_2$  and  $CO_2$  fractions were recorded with an open-circuit spirometry system (CPX/D Cardiopulmonary Exercise System, Medical Graphics, Minneapolis, MN, USA). Temperature, barometric pressure, and relative humidity, used for determination of end-tidal  $PO_2$  and  $PCO_2$  ( $PET_{O_2}/PET_{CO_2}$ ) and pulmonary gas exchange, were measured in the laboratory just prior to each experimental session. Arterial hemoglobin oxygen saturation ( $SA_{O_2}$ ) was continuously recorded with a finger pulse oximeter (Biox 3700e, Ohmeda, Madison, WI, USA), with the averaging function set at 3 s. Heart rate and arterial blood pressure were continuously recorded with a photoplethysmometer with the cuff attached on the left middle finger (Finapres 2300, Ohmeda, Madison, WI, USA). The Finapres has been shown to accurately record changes in mean arterial blood pressure during apnea (Parati et al., 1989). The relative skin capillary blood flow in the left thumb was continuously recorded with a laser-Doppler flowmeter (Advanced Laser Flowmeter 21, Advance Company Ltd., Tokyo, Japan). In the first eleven subjects, blood samples for determination of the plasma lactate concentration were taken from a venous catheter inserted in the right arm. Blood samples were collected 1 min before, as well as 1, 2, 3, and 4 min after the end of each sub-maximal duration

apnea and kept on ice until cold-centrifuged within 60 min. The separated plasma was then stored at  $-18^{\circ}\text{C}$  until it was analyzed (Modular P, Roche Diagnostics/Hitachi Instruments, Inc.).

#### 2.4. Data analysis

The control, eupneic alveolar gas exchange was determined for the last minute prior to each apnea. Pulmonary  $\text{O}_2$  and  $\text{CO}_2$  exchange and the respiratory exchange ratio ( $R$ ) during apneas were calculated from the differences between volumes of  $\text{O}_2$  and  $\text{CO}_2$  in the lungs at the beginning of and at the end of apnea (for details see (Andersson et al., 2004)).

Control values for  $\text{SaO}_2$ , heart rate, skin blood flow, and mean arterial blood pressure (MAP) were calculated as averages from the period 90–30 s before each apnea. For each apnea, an average value of  $\text{SaO}_2$  was calculated for the 10-s period ending with the nadir  $\text{SaO}_2$ -value. For apneic values of heart rate, skin blood flow, and MAP, averages were calculated for the entire apneic period except the initial 30 s when the diving response is established (Jung and Stolle, 1981; Schagatay et al., 1999). The rate-pressure product (heart rate  $\times$  systolic arterial blood pressure) during the control and apneic periods was calculated for an estimate of the myocardial  $\text{O}_2$  demand (McArdle et al., 1996).

For each subject, an individual mean value from the two sub-maximal duration apneas in each condition was calculated for all parameters, and a group mean (S.E.) of the 16 individual means was calculated. Absolute apneic values from apneas in air and apneas with face immersion were compared to the eupneic control values, and values from apneas in air were compared to the values from apneas with face immersion. All comparisons were done using paired Students  $t$ -test with Bonferroni correction for multiple comparisons. The level for accepting significance was  $P < 0.05$ .

### 3. Results

Reported data are group means (S.E.) from 16 subjects, except for the plasma lactate concentration values, which were obtained from 11 subjects.

#### 3.1. Breath-holding times

The maximum breath-holding time was 184 (9) s (range: 135–243 s). Based on the maximum breath-holding time, the sub-maximal breath-holding time was determined to be 145 (7) s (range: 120–210 s), and all subjects were able to complete every apnea to the predetermined time.

#### 3.2. Pulmonary gas exchange

The  $\text{PETO}_2$  and  $\text{PETCO}_2$  values obtained in the last expiration prior to apneas and in the first post-apneic expiration are shown together with the  $\text{SaO}_2$  in Table 1. The  $\text{PETO}_2$  and  $\text{SaO}_2$  decreased more during apneas in air than during apneas with face immersion. There was no difference in the increase in  $\text{PETCO}_2$  between apneas in air and apneas with face immersion.

Table 1

End-tidal  $\text{PO}_2$ ,  $\text{PCO}_2$ , and  $\text{SaO}_2$  before (pre) and after apneas in air (post-A) and apneas with face immersion (post-AFI)

|          | $\text{PETO}_2$ (mmHg)    | $\text{PETCO}_2$ (mmHg) | $\text{SaO}_2$ (%)       |
|----------|---------------------------|-------------------------|--------------------------|
| Pre      | 115 (1)                   | 34 (1)                  | 98 (0.2)                 |
| Post-A   | 57 (3) <sup>†††</sup>     | 51 (1) <sup>†††</sup>   | 90 (2) <sup>††</sup>     |
| Post-AFI | 61 (3) <sup>†††,***</sup> | 51 (1) <sup>†††</sup>   | 92 (2) <sup>††, **</sup> |

Values are means (S.E.) from 16 subjects. <sup>††</sup> $P < 0.01$  and <sup>†††</sup> $P < 0.001$  between pre and post; <sup>\*\*</sup> $P < 0.001$  and <sup>\*\*\*</sup> $P < 0.001$  between post-A and post-AFI.

The total pulmonary  $\text{O}_2$  uptake during the apneas was reduced compared to the eupneic control level, and the  $\text{O}_2$  uptake was lower during apneas with face immersion than during apneas in air (Fig. 1). During apneas in air and apneas with face immersion, the  $\text{O}_2$  uptake had been reduced from the control level by 19 (4) and 23 (4)%, respectively. Also the blood-to-lung  $\text{CO}_2$  elimination during apneas was reduced compared to the eupneic control level (Fig. 1). There was no difference in  $\text{CO}_2$  elimination between apneas in air and apneas with face immersion. During both apneas in air and apneas with face immersion the  $\text{CO}_2$  elimination was reduced from the control level by 60 (4)%. With the changes in  $\text{CO}_2$  elimination and  $\text{O}_2$  uptake, the  $R$  was reduced during apneas ( $P < 0.001$ ) compared to the control level, 0.95 (0.02). During apneas in air the  $R$  was 0.47 (0.02) and during apneas with face immersion 0.49 (0.02) ( $P < 0.01$  between apneas in air and apneas with face immersion).

#### 3.3. Plasma lactate

In all samples the venous plasma lactate concentration was 1.1 (0.1) mM, i.e., it was unchanged during recovery from apneas compared to the control, pre-apnea level.

#### 3.4. Cardiovascular responses

During the control period before apneas, the heart rate was 73 (4) bpm and the mean arterial blood pressure (MAP) was 125 (3) mmHg. During apneas in air, the heart rate was reduced from eupneic control by 10 (2)% ( $P < 0.05$ ), while the MAP increased by 14 (2)% ( $P < 0.001$ ). The heart rate reduction from control was augmented by face immersion in cold water to 20

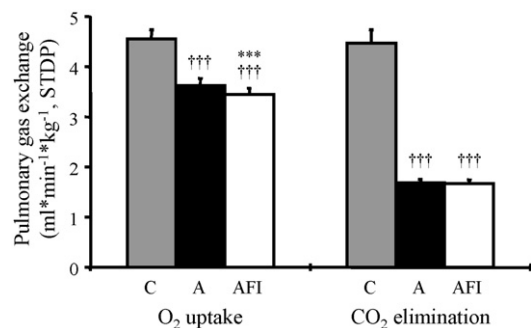


Fig. 1. Time-averaged pulmonary  $\text{O}_2$  uptake and  $\text{CO}_2$  elimination during eupneic control and during apneas in air and apneas with face immersion (C, A, and AFI, respectively). Values are means (S.E.) from 16 subjects. <sup>†††</sup> $P < 0.001$  compared to control; <sup>\*\*\*</sup> $P < 0.001$  between A and AFI.

(2%) ( $P < 0.001$  between apneas in air and apneas with face immersion), but the MAP increased to the same extent, 16 (2)%, as during apneas in air. The rate-pressure product was reduced by 10 (3)% from the eupneic control level during apneas with face immersion ( $P < 0.05$  compared to control), but there was no change from control during apneas in air. A tendency for a reduced skin blood flow was observed during apneas in air (–23 (7)%;  $P = 0.07$  compared to control). During apneas with face immersion, the skin blood flow was reduced by 32 (5)% from the control level ( $P < 0.05$ ).

#### 4. Discussion

This study shows that an augmented diving response, initiated during apnea with face immersion in resting humans, is associated with a reduced pulmonary  $O_2$  uptake compared to during apnea alone. This effect of the diving response on the  $O_2$  uptake in resting humans has not been demonstrated previously. The rate of depletion of the lung  $O_2$  store is temporarily reduced because of the cardiovascular effects, most likely including reduced cardiac output and a redistribution of blood flow, preserving the lung  $O_2$  store for use by primarily vital organs. Thus, it can be concluded that the diving response has an  $O_2$ -conserving effect in resting humans, in accordance with earlier observations (Ferretti et al., 1991; Linér and Linnarsson, 1994; Andersson and Schagatay, 1998).

With a preserved lung  $O_2$  store during apnea, the arterial  $PO_2$  would be falling at a slower rate, and thus the breath-holding time could be prolonged without threatening the function of the heart and the brain. In accordance with this, Schagatay and Andersson (1998) observed longer breath-holding times accompanied by an augmented diving response during cold-water face immersion apneas compared to during apneas in air in trained breath-hold divers. However, this was not observed in untrained subjects in which psychological rather than physiological factors determined apneic duration (Schagatay and Andersson, 1998). Considering less psychologically susceptible variables than the breath-holding time, it has been shown in both trained breath-hold divers and novice subjects that an augmented diving response reduces the rate of arterial hemoglobin desaturation during both rest and exercise (Andersson and Schagatay, 1998; Lindholm et al., 1999; Andersson et al., 2002; Stewart et al., 2005). Furthermore, it has been shown that the cardiovascular responses reduce the pulmonary  $O_2$  uptake during apnea in exercising subjects (Lindholm and Linnarsson, 2002; Andersson et al., 2004). The effect of the human diving response on pulmonary gas exchange shown in exercising subjects is further substantiated, and extended also to the resting condition, by the results of the present study. Many apneic conditions, e.g. sleep apnea and competitive “static apnea”, are associated with no or very low levels of physical activity. In this context it is noteworthy that the current world record for breath holding exceeds 9 min. Larger than normal  $O_2$  stores at the beginning of apnea, due to hyperventilation and large lung volumes (Muth et al., 2003; Lemaitre et al., 2005; Lindholm and Lundgren, 2006), erythrocyte release from the contracting spleen (Schagatay et al., 2001; Bakovic et al., 2003), and a reduction in the rate of depletion of the lung  $O_2$

store due to the diving response as shown in the present study, help to explain these remarkable achievements.

The mechanisms behind the temporarily reduced  $O_2$  uptake during apnea depend on a reduction in cardiac output and a redistribution of blood flow. Linér and Linnarsson (1994) observed a reduced pulmonary  $O_2$  uptake during apneas compared to the eupneic control in resting subjects. The reduced  $O_2$  uptake was attributed to a decrease in cardiac output together with changes in the distribution of blood flow and venous blood volume during apneas. With the reduced  $O_2$  delivery to peripheral tissues because of the peripheral vasoconstriction and decrease in cardiac output, the lung  $O_2$  store will be preserved at the expense of the peripheral venous  $O_2$  stores (Linér, 1994; Andersson et al., 2004). In accordance with this, using near-infrared spectroscopy, an earlier skeletal muscle desaturation compared to arterial and cerebral desaturation have been observed (Valic et al., 2006; Palada et al., 2007). Taken together, these previous results along with the findings of the present study supports the notion that the human diving response has an oxygen-conserving effect. As the cardiovascular changes of the diving response are enhanced by face immersion, the reduction in  $O_2$  uptake during apnea observed in the present study was augmented during apnea with face immersion.

The blood-to-alveolar  $CO_2$  elimination was markedly reduced during apnea compared to the eupneic control level, which has previously been observed in both resting and exercising subjects (Lanphier and Rahn, 1963; Hong et al., 1971; Lindholm and Linnarsson, 2002). It is explained by a combination of the Haldane effect and an elevation of alveolar  $PCO_2$  by lung volume shrinkage causing the alveolar–arterial  $PCO_2$  to increase faster than the mixed venous  $PCO_2$  (Lanphier and Rahn, 1963). As the  $PCO_2$  gradient between mixed venous blood and alveoli changes,  $CO_2$  exchange between blood and lungs is reduced and may even be reversed. The decreased  $CO_2$  elimination causes the marked reduction in the respiratory exchange ratio in the present study.

There was no increase in venous plasma lactate concentration after apneas. This was contrary to our hypothesis and unexpected, as previous studies have shown increased blood lactate levels after apneas in resting subjects (Olsen et al., 1962; Wolf et al., 1965; Hong et al., 1971; Joulia et al., 2003). Furthermore, increases in plasma lactate concentration have been observed after apneas combined with exercise of low intensities (Ferretti et al., 1991; Joulia et al., 2003; Andersson et al., 2004). It is possible that we may have missed a peak in plasma lactate concentration as we only collected blood samples at 1-min intervals after the end of apneas. Alternatively, the lack of an increase in plasma lactate concentration could relate to the fact that the majority of our subjects were trained breath-hold divers, as Joulia et al. (2003) found that a 3-month apnea-training programme attenuated the increase in blood lactic acid concentration observed after apnea. Whether this observation reflects a reduced lactic acid production, increased catabolism, or a reduced release from the tissues to the blood (due to the peripheral vasoconstriction) remains to be elucidated.

In conclusion, the diving response is shown to at least temporarily reduce the pulmonary  $O_2$  uptake during apneas



compared to eupneic control in resting humans. Furthermore, we show that an augmented diving response during apneas with face immersion is associated with a lower pulmonary O<sub>2</sub> uptake compared to during apneas in air. This is most likely caused by reductions in cardiac output and peripheral blood flow during apnea, indicated by the observed cardiovascular changes during apneas in the present study. The peripheral, predominantly venous, O<sub>2</sub> stores are probably relatively more reduced, while the lung O<sub>2</sub> store is preserved with a more pronounced diving response. Therefore, the development of hypoxic levels that threaten the function of the heart and the brain during apnea will be delayed with a more pronounced diving response, allowing a prolongation of apneic time.

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